

Final Report - Task 4: Data Storytelling & Statistical Validation

Project Objective

Analyze sales performance using Power BI dashboards, derive business insights, and validate findings using statistical methods.

Dataset Overview

The project uses a sales transaction dataset containing order, product, customer, regional, sales, profit, and quantity information. Data was cleaned and transformed before analysis.

Dashboard Summary

The dashboard includes KPI Cards, Sales Funnel, Category Analysis, Regional Analysis, Monthly Sales & Profit Trends, Top Products, Top Customers, and interactive slicers.

Key Insights

- KPI monitoring provides a quick view of business performance.
- Category and regional analysis identify high- and low-performing segments.
- Monthly trends support forecasting and inventory planning.
- Top products contribute significantly to revenue.
- Funnel analysis highlights valuable orders.

Statistical Validation

A Pearson correlation framework was proposed to validate the relationship between Sales and Profit at a 95% confidence level.

Business Recommendations

- Increase focus on high-performing products.
- Improve sales strategies in weaker regions.
- Monitor KPIs regularly.
- Use statistical analysis to support business decisions.

Conclusion

Combining Power BI dashboards with statistical validation enables reliable, data-driven decision-making and improves overall business performance.